# Grade 11 – Term 1
# C2 Practice Activity 2 with Solutions

## Contents



## Evaluated criterion

- C2: Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.

# 1 Introduction

**Problems:**



The central skill in this activity is constructing a continuous probability model from a situation, rather than merely evaluating a supplied formula. For every problem, use the same eight-step routine: (1) read the narrative and define the random variable; (2) determine the **support** and geometric shape; (3) translate verbal relationships into heights, ratios, slopes, or breakpoints; (4) use total area = 1 to determine all unknown quantities; (5) write the complete pdf, including 0 outside its support; (6) mark the region requested by the probability; (7) calculate its area with rectangles, triangles, trapezoids, or a complement; and (8) state the result in context when useful. A density is a height, so it may exceed 1; it is the *area* that must total 1. Endpoints do not affect a continuous probability, for example $P(a < X < b) = P(a \le X \le b)$.

### 1.1 Uniform distribution

A uniform density has one constant height on support $[L, U]$:

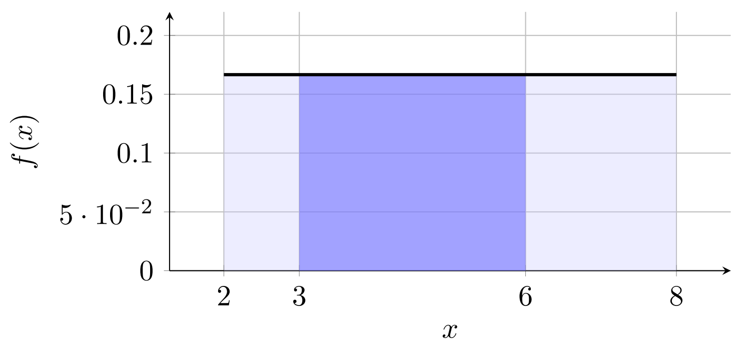
$$f_X(x) = \begin{cases} h, & L \le x \le U, \\ 0, & \text{otherwise.} \end{cases} \qquad (U - L)h = 1, \quad h = \frac{1}{U - L}.$$

Thus every probability is a rectangle: $P(a \le X \le b) = (b - a)h$ for $L \le a < b \le U$.

**Worked example.** A delivery time $T$ is equally plausible from 2 to 8 minutes. The support is $[2, 8]$ and the whole rectangle has width 6 and height $h$, so $6h = 1$ and $h = 1/6$. For $3 \le T \le 6$, the relevant rectangle has width $6 - 3 = 3$ and height $1/6$:

$$P(3 \le T \le 6) = A_{\text{rectangle}} = 3\left(\frac{1}{6}\right) = \frac{1}{2}.$$

There is a 50% chance that delivery takes from 3 to 6 minutes.




### 1.2 Linear distribution

A single linear density has $f_X(x) = mx + d$ throughout one support $[L, U]$ (and 0 outside), with nonnegative endpoint heights $h_L = f(L)$ and $h_U = f(U)$. Its whole area is the trapezoid

$$1 = \frac{1}{2}(h_L + h_U)(U - L).$$

An interval $[a, b]$ is also a trapezoid of width $b - a$ and parallel heights $f(a), f(b)$. A triangle is only the special case in which one endpoint height is zero.

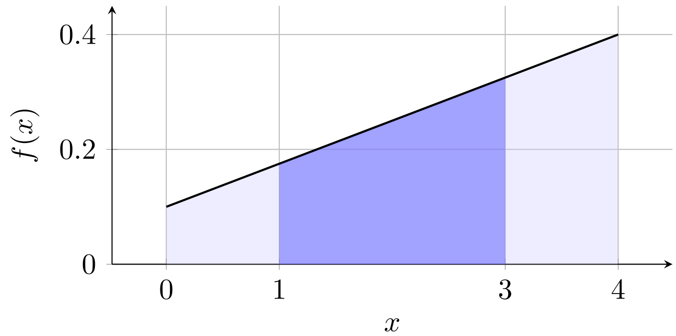
**Worked example.** A shop's daily price adjustment $X$ (in dollars) lies between 0 and 4. Its likelihood density increases linearly, and the density at 4 is four times that at 0. Translating the description, let the endpoint heights be $h$ and $4h$. The whole trapezoid gives

$$1 = \tfrac{1}{2}(h + 4h)4 = 10h, \qquad h = 0.1.$$

The line through $(0, 0.1)$ and $(4, 0.4)$ is $f_X(x) = 0.075x + 0.1$ on the support (and 0 otherwise). For $1 \le X \le 3$, $f(1) = 0.175$ and $f(3) = 0.325$. The required figure is a trapezoid of width 2 with those parallel heights:

$$P(1 \le X \le 3) = \tfrac{1}{2}(0.175 + 0.325)(2) = 0.5.$$

Both support endpoints have positive density, so this is genuinely linear, not triangular.




## 1.3 Triangular distribution

A triangular density has support $[L, U]$, height 0 at both ends, and mode $M$ $(L < M < U)$ with height $H$. It rises linearly on $[L, M]$ and falls linearly on $[M, U]$. Since the whole figure has base $U - L$,

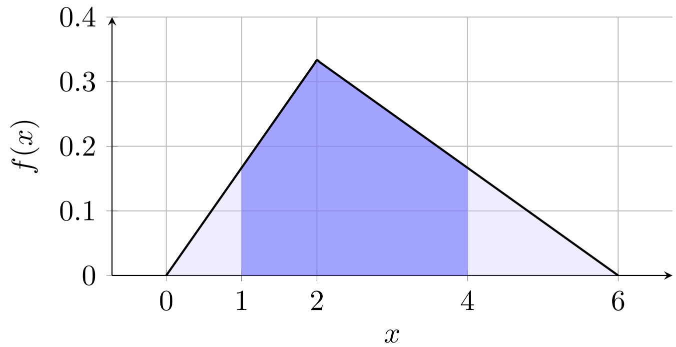
$$1 = \tfrac{1}{2}(U - L)H, \qquad H = \frac{2}{U - L}.$$

Heights inside either sloping side are found by proportional similar triangles.

**Worked example.** A project's cost overrun $C$ (thousand dollars) has minimum 0, most likely value 2, and maximum 6. Its support is $[0, 6]$, mode is 2, and $\frac{1}{2}(6)H = 1$, so $H = 1/3$. To find $P(1 \le C \le 4)$, subtract two tail triangles. At $C = 1$ the height is $1/6$; at $C = 4$ it is also $1/6$. The left tail has base 1, height $1/6$; the right tail has base 2, height $1/6$:

$$P(1 \le C \le 4) = 1 - \tfrac{1}{2}(1)(\tfrac{1}{6}) - \tfrac{1}{2}(2)(\tfrac{1}{6}) = 1 - \tfrac{1}{12} - \tfrac{1}{6} = \tfrac{3}{4}.$$

The region crosses the mode, but subtraction avoids splitting it into two trapezoids.



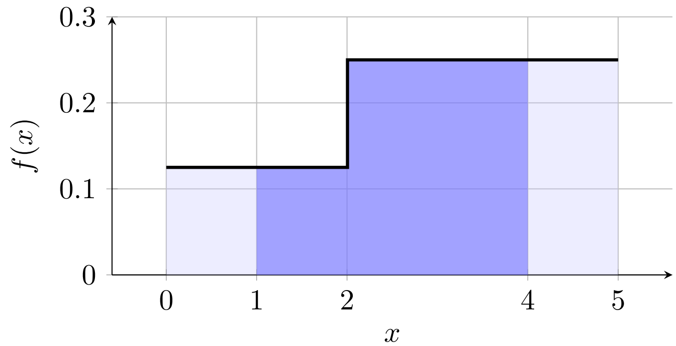

## 1.4 Piecewise uniform distribution

Here the density is constant *within* each interval but may jump at a breakpoint. Normalisation is the sum of rectangles: $1 = \sum(\text{segment width})(\text{segment height})$.

**Worked example.** A warehouse waiting time $W$ ranges from 0 to 5 minutes. Waiting times are equally dense within each of two regimes, 0–2 and 2–5, but the later regime has twice the density of the earlier one. If the first height is $k$, the narrative makes the second $2k$. A width-2, height-$k$ rectangle plus a width-3, height-$2k$ rectangle must have area 1:

$2k+6k=1$, so $k=1/8$. Thus the pdf has heights $1/8$ and $1/4$ on those respective regimes, and is 0 otherwise. The event $1 \le W \le 4$ spans the jump and uses two rectangles:

$$P(1 \le W \le 4) = \underbrace{(2-1)(1/8)}_{A_1} + \underbrace{(4-2)(1/4)}_{A_2} = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}.$$




## 1.5 Piecewise linear distribution

A piecewise-linear density joins straight segments at stated breakpoints. Read the height at each endpoint and breakpoint, then treat each piece as a triangle, rectangle, or trapezoid. Continuity means adjacent pieces meet at the same height.

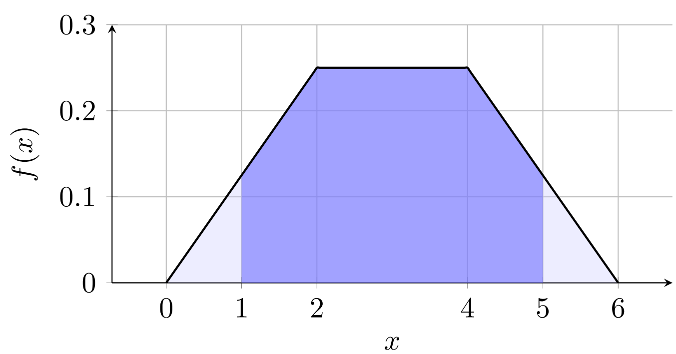
**Worked example.** Demand $D$ ranges from 0 to 6. Its density starts at zero, rises linearly to its maximum at 2, stays at that maximum until 4, and then falls linearly to zero at 6. Calling the maximum $H$ translates the story into a left triangle (base 2), rectangle (width 2), and right triangle (base 2):

$$1 = \tfrac{1}{2}(2)H + (2)H + \tfrac{1}{2}(2)H = 4H, \qquad H = \tfrac{1}{4}.$$

For $1 \le D \le 5$, subtract the two congruent tail triangles. By linearity, the heights at 1 and 5 are $H/2 = 1/8$:

$$P(1 \le D \le 5) = 1 - \tfrac{1}{2}(1)(\tfrac{1}{8}) - \tfrac{1}{2}(1)(\tfrac{1}{8}) = \frac{7}{8}.$$

This reusable approach identifies all three geometric pieces and uses subtraction from total area 1.



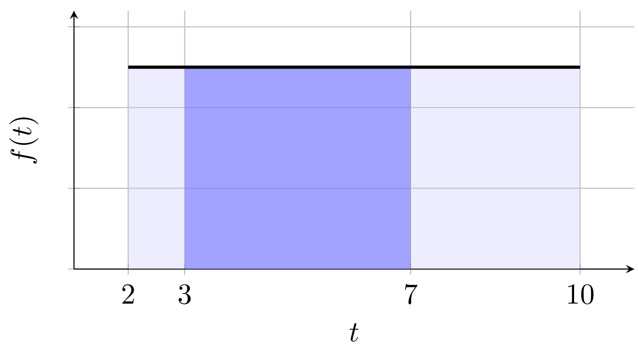
# 2 Uniform Distribution

**Problems:**



## 2.1 Microcredit disbursement window

A Bogotá fintech releases approved working-capital loans after fraud and bank-account checks. Let $T$ be the number of hours from approval to disbursement; differing bank response times make it continuous. Automatic cases never finish before 2 hours or after 10 hours, and operational data show that every time in that window is equally plausible. A bakery owner needs the money within 7 hours to meet a supplier's dispatch deadline. The fintech separately studies the central 3–7 hour band when assigning automated-processing capacity.




**Questions/Tasks.**

**C2.1** Using the situation, construct the complete pdf of $T$: identify its support, determine the unknown constant height from total area 1, and include 0 outside the support.

**C2.2** Calculate the deadline risk $P(T > 7)$ geometrically.

**C2.3** Calculate $P(3 \leq T \leq 7)$ from the appropriate geometric area.

**C2.1 - Construct the pdf from the situation.**

The narrative defines $T$ as disbursement time, bounds it by 2 and 10, and says all times in that window are equally plausible. Thus the support is $[2, 10]$ and its density is a rectangle of unknown height $h$. Normalisation gives

$$1 = A_{\text{rectangle}} = (10 - 2)h = 8h, \qquad h = \frac{1}{8}.$$

Therefore the complete model is

$$f_T(t) = \begin{cases} \frac{1}{8}, & 2 \leq t \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

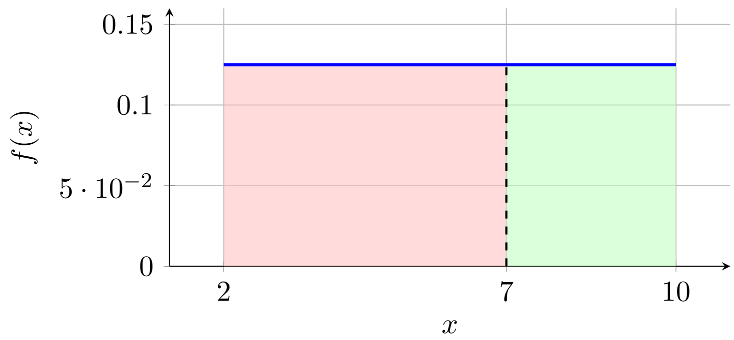
Its area is $8(1/8) = 1$, as required.

**C2.2 - Calculate the tail probability $P(T > 7)$.**

The right tail is a rectangle of width $10 - 7 = 3$ and height $1/8$:

$$P(T > 7) = A_{\text{rectangle}} = (10 - 7)\frac{1}{8} = \frac{3}{8} = 37.5\%.$$

Thus the bakery has a 37.5% modeled risk of missing its deadline.




**C2.3 - Calculate the interval probability $P(3 \leq T \leq 7)$.**

The central band is a rectangle of width $7 - 3 = 4$:

$$P(3 \leq T \leq 7) = (7-3)\frac{1}{8} = \frac{1}{2}.$$

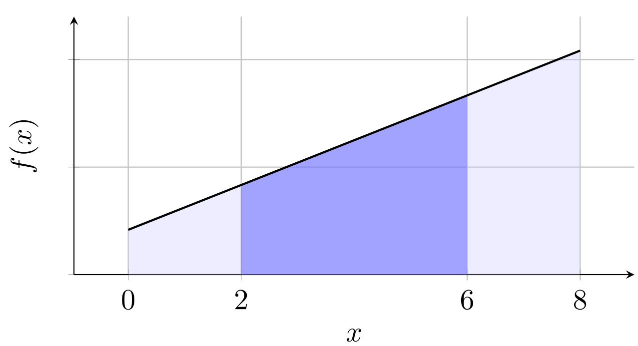
Hence half of comparable disbursements occupy this capacity-planning band.

# 3 Linear Distribution

| **Problems:** |
| --- |
| 3.1. Coffee auction premium |

## 3.1 Coffee auction premium

A Bogotá specialty-coffee roaster buys small lots from Colombian producers. Let $X$ be the continuously measured premium, in thousands of pesos per kilogram, above the reference price. Auction rules keep it between 0 and 8. Buyer competition makes its likelihood density increase at a constant rate across that range, and market records indicate that the density at the upper limit is five times the density at the lower limit. Premiums above 6 thousand pesos threaten margins, while the 2–6 band can be absorbed without changing menu prices.




**Questions/Tasks.**

**C2.1** Construct the complete pdf of $X$ from the narrative: determine both endpoint heights, derive the line, and include 0 outside the support.

**C2.2** Calculate the margin-risk tail $P(X > 6)$ geometrically.

**C2.3** Calculate $P(2 \le X \le 6)$ using the appropriate geometric area.

**C2.1 - Construct the pdf from the situation.**

The auction bounds give support $[0, 8]$. "Increases at a constant rate" gives one straight line, and the endpoint relationship gives heights $h$ and $5h$, both positive. The full trapezoid must have area 1:

$$1 = \frac{1}{2}(8)(h + 5h) = 24h, \qquad h = \frac{1}{24}, \quad 5h = \frac{5}{24}.$$

The line through $(0, 1/24)$ and $(8, 5/24)$ has slope $(5/24 - 1/24)/8 = 1/48$ and intercept $1/24 = 2/48$. Hence

$$f_X(x) = \begin{cases} \dfrac{x+2}{48}, & 0 \le x \le 8, \\ 0, & \text{otherwise.} \end{cases}$$

Both endpoint heights are positive, so the model is linear rather than triangular, and its trapezoidal area is 1.

**C2.2 - Calculate the tail probability $P(X > 6)$.**

The heights are $f_X(6) = 1/6$ and $f_X(8) = 5/24$. The right tail is a width-2 trapezoid:

$$P(X > 6) = \frac{1}{2}(8 - 6)\left(\frac{1}{6} + \frac{5}{24}\right) = \frac{3}{8}.$$

Thus 37.5% of modeled auctions exceed the manager's margin threshold.

**C2.3 - Calculate the interval probability $P(2 \le X \le 6)$.**

Since $f_X(2) = 1/12$ and $f_X(6) = 1/6$, the interval is a nonconstant trapezoid of width 4:

$$P(2 \le X \le 6) = \frac{1}{2}(6 - 2)\left(\frac{1}{12} + \frac{1}{6}\right) = \frac{1}{2}.$$

Thus the absorbable premium band occurs in half of the modeled auctions.

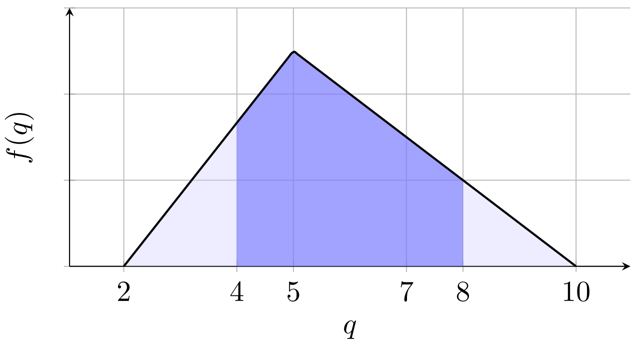
# 4 Triangular Distribution

**Problems:**



## 4.1 Last-mile order volume

A Bogotá last-mile cooperative contracts motorcycles before a national online sales event. Let $Q$ be continuously modeled next-day order volume in thousands of orders. Forecasts rule out fewer than 2 or more than 10 thousand orders; among feasible volumes, 5 thousand is most likely. The likelihood rises steadily from zero at the minimum to that single peak and then falls steadily to zero at the maximum. Demand above 7 thousand triggers an expensive external-courier contract, while 4–8 thousand is the range used to evaluate the regular fleet.




**Questions/Tasks.**

**C2.1** Construct the complete pdf of $Q$ from the forecast: determine the peak height and both linear equations, and include 0 outside the support.

**C2.2** Calculate the external-contract tail $P(Q > 7)$ geometrically.

**C2.3** Calculate $P(4 \leq Q \leq 8)$, an interval crossing the most likely value.

**C2.1 - Construct the pdf from the situation.**

The minimum, most likely value, and maximum translate to $L = 2$, $M = 5$, and $U = 10$. Rising from and falling to zero makes one triangle. If its peak is $H$, normalisation gives

$$1 = \frac{1}{2}(10 - 2)H, \qquad H = \frac{1}{4}.$$

The rising slope is $(1/4)/(5 - 2) = 1/12$; the falling slope is $-(1/4)/(10 - 5) = -1/20$. Using the zero-height endpoints,

$$f_Q(q) = \begin{cases} \dfrac{q-2}{12}, & 2 \leq q \leq 5, \\ \dfrac{10-q}{20}, & 5 < q \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

The pieces meet at height 1/4 and the total triangular area is 1.

**C2.2 - Calculate the tail probability $P(Q > 7)$.**

On the descending side, $f_Q(7) = 3/20$. The right tail is a triangle of base $10 - 7 = 3$ and height $3/20$:

$$P(Q > 7) = \frac{1}{2}(3)\frac{3}{20} = \frac{9}{40} = 22.5\%.$$

This is the modeled chance that external couriers are required.

**C2.3 - Calculate the interval probability $P(4 \leq Q \leq 8)$.**

The interval crosses the mode. A clean geometric decomposition subtracts its two outside triangles. Since $f_Q(4) = 1/6$ and $f_Q(8) = 1/10$,

$$P(4 \leq Q \leq 8) = 1 - \underbrace{\frac{1}{2}(4-2)\frac{1}{6}}_{\text{left triangle}} - \underbrace{\frac{1}{2}(10-8)\frac{1}{10}}_{\text{right triangle}} = \frac{11}{15}.$$

Thus the regular-fleet range covers about 73.3% of modeled days.

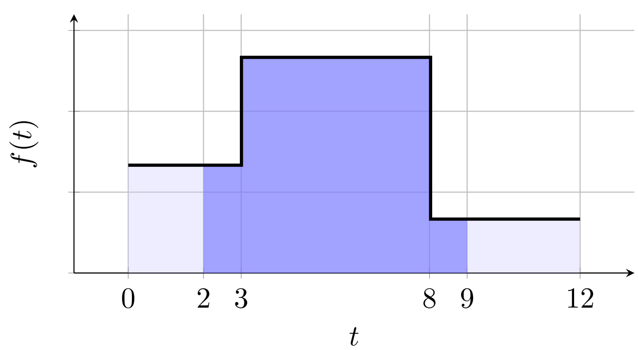
# 5 Piecewise Uniform Distribution

**Problems:**



## 5.1 Merchant reconciliation time

A Colombian payments processor reconciles QR-code transactions before releasing merchant balances. Let $T$ be the continuously measured time in minutes. No automatic reconciliation takes less than 0 or more than 12 minutes. The workflow has three regimes: initial validation from 0 to 3, ledger matching from 3 to 8, and exception closing from 8 to 12. Times are equally dense within each regime. Ledger-matching observations occur at twice the density of initial-validation observations, while exception-closing observations occur at half the density of initial-validation observations. Times above 8 require the slower exception queue; the processor also evaluates a 2–9 minute service window that crosses all three regimes.




**Questions/Tasks.**

**C2.1** Construct the complete pdf of $T$ from the three regimes: translate their relative densities, normalize the heights, and include 0 outside the support.

**C2.2** Calculate the exception-queue tail $P(T > 8)$ geometrically.

**C2.3** Calculate $P(2 \leq T \leq 9)$ by decomposing the interval at the breakpoints.

**C2.1 - Construct the pdf from the situation.**

The stated bounds give support $[0, 12]$, with breakpoints 3 and 8. Let the initial-validation height be $k$. The verbal ratios translate the next two heights to $2k$ and $k/2$. The three rectangles must total 1:

$$1 = 3k + (8-3)(2k) + (12-8)\frac{k}{2} = 15k, \qquad k = \frac{1}{15}.$$

Therefore

$$f_T(t) = \begin{cases} \frac{1}{15}, & 0 \le t < 3, \\ \frac{2}{15}, & 3 \le t < 8, \\ \frac{1}{30}, & 8 \le t \le 12, \\ 0, & \text{otherwise.} \end{cases}$$

All heights are nonnegative and the component areas sum to 1.

**C2.2 - Calculate the tail probability $P(T > 8)$.**

The exception tail is a width-4, height-1/30 rectangle:

$$P(T > 8) = (12 - 8)\frac{1}{30} = \frac{2}{15} \approx 13.3\%.$$

The endpoint does not affect a continuous probability.

**C2.3 - Calculate the interval probability $P(2 \le T \le 9)$.**

Split at both regime boundaries and add three rectangles:

$$\begin{aligned} P(2 \le T \le 9) &= \underbrace{(3-2)\frac{1}{15}}_{[2,3)} + \underbrace{(8-3)\frac{2}{15}}_{[3,8)} + \underbrace{(9-8)\frac{1}{30}}_{[8,9]} \\ &= \frac{23}{30} \approx 76.7\%. \end{aligned}$$

This is the modeled proportion inside the processor's service window.

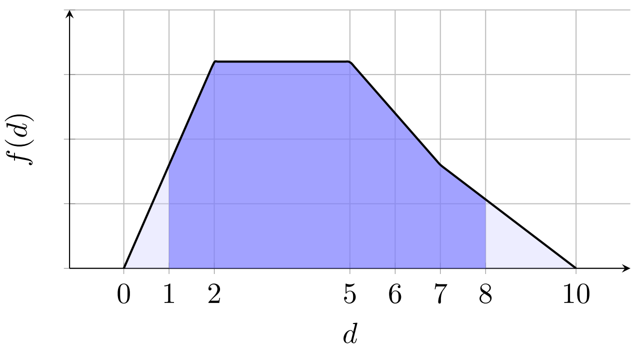
# 6 Piecewise Linear Distribution

**Problems:**



## 6.1 Weekly payout liquidity

A Bogotá marketplace holds cash for Friday seller payouts, refunds, and chargebacks. Let $D$ be continuously measured required liquidity in billions of pesos. Requirements range from 0 to 10. Their density begins at zero, rises at a constant rate to its maximum when requirements reach 2, and stays at that maximum through 5. It then falls at a constant rate until, at 7, it is half the maximum; from there it falls at a constant rate to zero at 10. Requirements above 6 trigger an expensive backup credit line. The 1–8 billion range is used to assess how often the standard treasury plan covers the week.




**Questions/Tasks.**

**C2.1** Construct the complete pdf of $D$ from the narrative: determine the peak, derive every segment equation, and include 0 outside the support.

**C2.2** Calculate the backup-credit tail $P(D > 6)$ by decomposing it at the structural change.

**C2.3** Calculate $P(1 \le D \le 8)$, an interval spanning all structural pieces.

### C2.1 - Construct the pdf from the situation.

The story gives support $[0, 10]$ and breakpoints $2, 5, 7$. If the maximum is $H$, the corresponding heights are $0, H, H, H/2, 0$. Thus the geometry is a base-2 triangle, width-3 rectangle, width-2 trapezoid, and base-3 triangle. Normalisation gives

$$1 = \frac{1}{2}(2)H + 3H + \frac{1}{2}(2)\left(H + \frac{H}{2}\right) + \frac{1}{2}(3)\frac{H}{2} = \frac{25}{4}H, \qquad H = \frac{4}{25}.$$

Deriving each line from its two endpoint values gives slopes $2/25$, $0$, $-1/25$, and $-2/75$, respectively. Therefore

$$f_D(d) = \begin{cases} \dfrac{2d}{25}, & 0 \le d \le 2, \\ \dfrac{4}{25}, & 2 < d \le 5, \\ \dfrac{9-d}{25}, & 5 < d \le 7, \\ \dfrac{2(10-d)}{75}, & 7 < d \le 10, \\ 0, & \text{otherwise.} \end{cases}$$

Adjacent equations agree at every breakpoint, and the geometric areas total 1.

### C2.2 - Calculate the tail probability $P(D > 6)$.

The tail begins within the declining 5–7 segment. Its heights at 6 and 7 are $3/25$ and $2/25$. From 6 to 7 is a trapezoid; from 7 to 10 is a triangle:

$$P(D > 6) = \underbrace{\frac{1}{2}(7-6)\left(\frac{3}{25} + \frac{2}{25}\right)}_{\text{trapezoid}} + \underbrace{\frac{1}{2}(10-7)\frac{2}{25}}_{\text{triangle}}$$
$$= \frac{1}{10} + \frac{3}{25} = \frac{11}{50} = 22\%.$$

This is the modeled frequency with which backup credit is triggered.

**C2.3 - Calculate the interval probability $P(1 \leq D \leq 8)$.**

Although the interval spans every structural piece, its complement consists of two simple tail triangles. Since $f_D(1) = 2/25$ and $f_D(8) = 4/75$,

$$\begin{aligned} P(1 \leq D \leq 8) &= 1 - \underbrace{\frac{1}{2}(1)\frac{2}{25}}_{[0,1]} - \underbrace{\frac{1}{2}(10-8)\frac{4}{75}}_{[8,10]} \\ &= 1 - \frac{1}{25} - \frac{4}{75} = \frac{68}{75} \approx 90.7\%. \end{aligned}$$

Thus the standard treasury range covers about 90.7% of modeled weeks.